



Agaram
TECHNOLOGIES

CASE STUDY

BIOPHARMACEUTICAL MANUFACTURING ORGANIZATION

A globally recognized biopharmaceutical manufacturer has always prioritized innovation and operational excellence. However, in their QC labs, manually updating the **Körber PAS-X Manufacturing Execution System (MES)** after offline data capture from analytical instruments led to batch operation delays and increased the risk of data errors and compliance issues.

To overcome this, the manufacturer organization had decided to implement Agaram's **Logilab SDMS**, a scientific data management and middleware platform that bridges laboratory instruments with enterprise systems.

Integrating tabletop lab instruments with PAS-X MES using Logilab SDMS marked a strategic transformation for the organization, enhancing operational precision, compliance, and scalability.

Integration of Clinical & Analytical instruments with MES for a Biopharma In-Process QC

Client: Global biopharmaceutical manufacturer

Partner: Agaram Technologies

Objective: Eliminate manual data entry from lab instruments and integrate seamlessly with PAS-X MES

Background & Challenge

Pharmaceutical labs often rely on standalone analytical instruments (pH meters, osmometers, etc.), creating a manual gap in data entry into MES systems. Our client, a global pharma leader, faced:

- **Manual transcription delays and errors**
- **No equipment usage visibility**
- **Compliance risks due to weak audit trails**
- **Complex integration across multiple instruments**

They needed a **compliant, scalable, and real-time integration** solution to digitize instrument data capture and automate workflows within PAS-X MES.

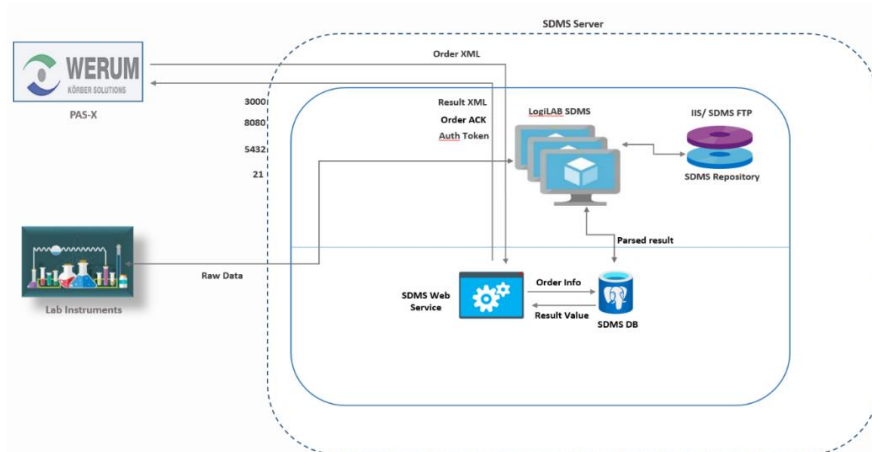
The Solution: Logilab (SDMS +IoT Connect Middleware) + PAS-X MES Integration

Agaram implemented its **Logilab SDMS** platform to serve as a secure, middleware layer between lab instruments and PAS-X MES. The project introduced a **real-time, metadata-driven integration** using REST APIs and a modular architecture.





The schematic would be as under



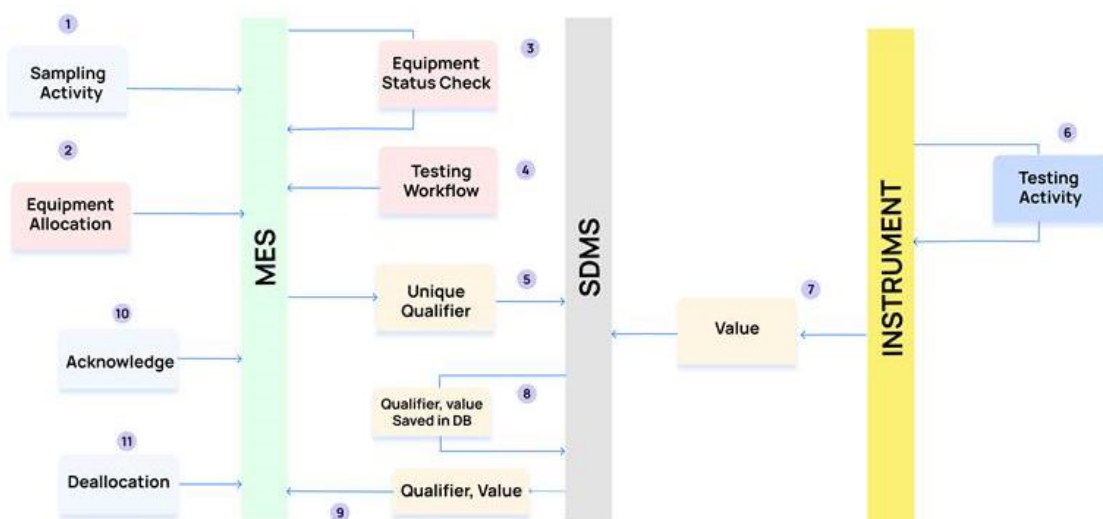
Key Capabilities Delivered:

- ✅ **Automated Data Capture:** From RS232/file-based instruments into SDMS
- ✅ **Real-Time RESTful API Integration:** SDMS pushes validated data directly into PAS-X MES
- ✅ **Auto-Locking of Instruments:** Instruments are locked/unlocked based on MES allocation — reducing errors and ensuring compliance
- ✅ **Regulatory-Ready Architecture:** Supports 21 CFR Part 11, ALCOA+, and GMP requirements
- ✅ **Scalable Design:** Built to scale from 1 to 50+ instruments with minimal configuration

How It Works

1. Instrument generates data →
2. MOXA device converts RS232 to TCP/IP →
3. Logilab SDMS parses, tags, and stores results →
4. REST API pushes data to PAS-X MES →
5. MES updates batch records; SDMS unlocks instrument

This integration removed manual steps and ensured **traceable, tamper-proof, and audit-ready workflows**.



INTEGRATION WORKFLOW



Outcomes & Benefits

- ↓ **XXXXXXXXXX% Manual Errors:** No more data re-entry or transcription mistakes
- ↑ **Speed:** Instant availability of results in MES accelerated batch release
- ↑ **Equipment Utilization:** Auto-locking logic ensured purposeful instrument use
- ↑ **Compliance Confidence:** original data & PDF archival, audit trails, and metadata tagging support ALCOA+
- ↑ **Future-Readiness:** REST API-based model simplifies future expansion

Why Agaram Technologies?

We specialize in **digitizing laboratory ecosystems** with a focus on compliance, connectivity, and usability. Our Logilab SDMS platform integrates with leading MES, LIMS, and ERP systems, empowering pharmaceutical companies to scale faster while staying audit ready.

Looking to digitize and automate lab operations in your pharma production environment?

Contact us at info@agaramtech.com to explore how we can support your transformation journey.

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